

Name: _____

STATISTICS, (HISTOGRAMS)

Date: _____

LO: I can draw and interpret histograms with equal and unequal class widths.

Histograms

A **histogram** displays continuous data using bars with no gaps. The **frequency density** is plotted on the y-axis when class widths are unequal.

Key formula: Frequency density = frequency $\div$ class width.
The area of each bar represents the frequency.

Quick examples

●	Freq density = $20 \div 5 = 4$	frequency 20, class width 5
●	Frequency = density $\times$ width	area of bar = frequency
●	No gaps between bars	data is continuous
●	Plot frequency on y-axis for unequal widths	must use frequency density

Key idea

Frequency density = frequency divided by class width. Area of bar = frequency.

Frequency density =

frequency $\div$ class width

Area of each bar gives the frequency for that class

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - calculating frequency density

Class 10-20 has frequency 30. Find the frequency density.

$$\text{Class width} = 20 - 10 = 10$$

$$\text{Frequency density} = 30 \div 10 = 3$$

Frequency density = 3

Divide the frequency by the class width.

Example 2 - finding frequency from a histogram

A bar has height (density) 4 and width 5. Find the frequency.

$$\begin{aligned} \text{Frequency} &= \text{density} \times \text{width} \\ &= 4 \times 5 \\ &= 20 \end{aligned}$$

Frequency = 20

The area of the bar gives the frequency.

Example 3 - unequal class widths

Classes 0-10 (freq 15), 10-30 (freq 40). Find densities.

$$0-10: \text{width } 10, \text{ density} = 15 \div 10 = 1.5$$

$$10-30: \text{width } 20, \text{ density} = 40 \div 20 = 2$$

Bar heights: 1.5 and 2

1.5 and 2

Each class has its own width; calculate density separately.

Example 4 - estimating from a histogram

Estimate how many values fall between 15 and 25 if the density is 3 across that range.

$$\text{Width of interest} = 25 - 15 = 10$$

$$\begin{aligned} \text{Frequency} &= \text{density} \times \text{width} = 3 \times 10 \\ &= 30 \text{ values} \end{aligned}$$

30 values

Use area to estimate frequencies within part of a class.

YOUR TURN – PRACTICE (deliberate practice)

1. Calculating frequency density

Find the frequency density for each class.

- a) Class 0-10, frequency 25
- b) Class 10-30, frequency 60
- c) Class 30-50, frequency 44
- d) Class 50-80, frequency 90

2. Finding frequency from density

Find the frequency (area of bar).

- a) Density 4, width 10
- b) Density 2.5, width 20
- c) Density 1.8, width 15
- d) Density 6, width 5

3. Completing a table

Find the missing values.

- a) Class 0-5, freq ?, density 4
- b) Class 5-15, freq 30, density ?
- c) Class 15-25, freq ?, density 2.5
- d) Class 25-50, freq 75, density ?

4. Interpreting histograms

Use the histogram data to answer.

- a) Total frequency if densities are 2, 3, 4 for widths 10, 10, 5?
- b) Which class has highest frequency: density 5 width 4, or density 3 width 8?
- c) Estimate median class if total freq = 60, first two classes have freq 15 and 25.
- d) Bar area = 36, width = 12. What is the height?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Frequency density = frequency ÷ class width ☐

Correct answer: _____

Reason: _____

b) The height of a histogram bar always shows the frequency ☐

Correct answer: _____

Reason: _____

c) Histogram bars have no gaps because data is continuous ☐

Correct answer: _____

Reason: _____

d) Wider bars always represent more data ☐

Correct answer: _____

Reason: _____

e) The area of each bar represents the frequency ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A histogram shows times (minutes): 0-5 (density 4), 5-15 (density 3), 15-30 (density 2), 30-60 (density 1). Find the total number of people surveyed. Estimate how many took between **10 and 20 minutes**.

Expression: _____

Simplified: _____

b) 80 students took a test. The results: 0-40 marks (density 0.5), 40-60 marks (density 2), 60-80 marks (density d), 80-100 marks (density 0.75). Find the value of d and estimate the median mark.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

